- Machine failure:
  - Inadequately guarded, height, strength etc.
  - Unsafely designed machines, tools etc.
  - Defective, rough, sharp, slippery, decayed machines
  - Unsafely arranged, congestion, blocked exits.
  - Unsafely clothed, no goggles, glares or masks, high heels etc.

- Types of Hazards:
  - Electrical hazards
  - Chemical hazards
  - Gas hazards
  - Dust explosion
  - Fire and Explosion hazards.

CHEMICAL HAZARDS

- Reasons like:
  - Wrong raw materials
  - Raw materials containing trace impurities
  - Changed operating conditions
  - Unanticipated time delays
  - Equipment failure
  - Wrong materials of construction

GAS HAZARDS

- Flammable liquids:

Vaporization causes air pollution

Ignition of vapor causing fire accidents and explosions

Further spreading leads to loss of life and property

- Location, storage and handling:

The properties of liquids are considered to design suitable methods for storage and handling

1. Flash point
2. Apparent ignition temperature
3. Explosive range
4. Vapour density

- ✓ Stored in isolated places.
- ✓ Using tanks constructed above ground and open place
- ✓ Methods of handling include pumping, gravity flow and compressed air.

CONTROL OF DUST EXPLOSION

Can be controlled at the source of generation and the methods include

1. Filtration
2. Inertial separation
3. Electrostatic precipitation

Fire and Explosion Hazards

What is the source?

- Spontaneous combustion
- Non-explosion-proof electrical equipment

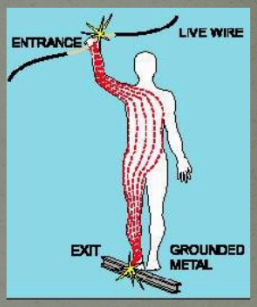
Fire occurs in the industry more frequently than an explosion, or toxic release, but the loss in terms of life is less...

Control Fire & Explosion

- Building construction
- Exit points
- Fire alarm equipment
- Fire Extinguishers
- Sprinkler systems
- Hydrants
- Safety and Fire protection organization

Shock Severity

- Severity of the shock depends on:
  - Path of current through the body
  - Amount of current flowing through the body (amps)
  - Duration of the shocking current through the body.
- LOW VOLTAGE DOES NOT MEAN LOW HAZARD



Electrical Injuries

There are three direct and two indirect types of electrical injuries:

- Direct:
  - Death due to electrical shock
  - Electrical shock
  - Burns
- Indirect:
  - Falls
  - Fire



What are the safety priorities?

Electrical hazards exist in almost every workplace. Common causes of electrocution are:

- making contact with overhead wires
- undertaking maintenance on live equipment
- working with damaged electrical equipment, such as extension leads, plugs and sockets
- using equipment affected by rain or water ingress

What should you do in an electrical emergency?

For low voltage electricity >50 V AC and 110 V DC

- remove the source of electricity supply
- call the emergency number on site

For high voltage electricity >1000 V

- call the emergency number for your site
- don't go near the casualty
- don't touch the casualty or try to free them with anything

Should you report electrical incidents?

- Electricity is invisible – this in itself makes it dangerous
- It has great potential to seriously injure or kill
- The company has a duty of care to its employees and contractors
- Everyone is exposed to electrical hazards, not just electricians
- Report all electrical shocks and near misses

All employees can be exposed to electrical Hazards. They should receive electrical hazard Training at the commencement of their Employment and REGULAR REFRESHER TRAINING.

Can you protect yourself from electricity?

- Don't wear metal objects
- Turn power off
- Wear appropriate clothing
- Don't touch live parts
- Don't install or repair electrical equipment
- Use qualified personnel
- Clean and dry leads and plugs before use



Special safety protection equipment:

- Head and Eyes: goggles, helmets, masks and hats.
- Hands, arms, legs and feet: rubber or canvas gloves, rubber boots, aprons, shoes and other clothing.
- Prevention of breathing of poisonous gases: respiration protective devices.

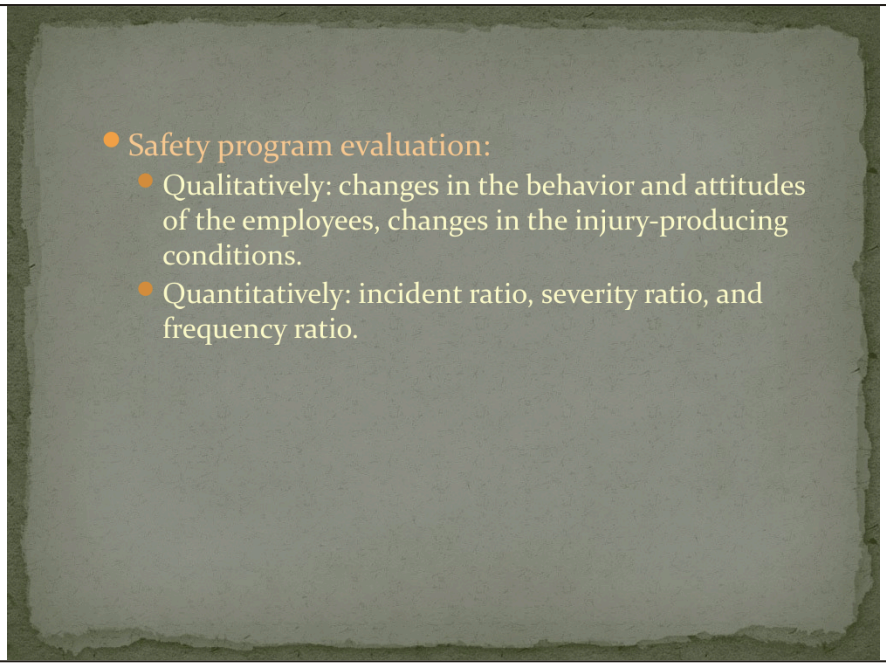
Safety programme

- Safety programme deals with the prevention of accidents and minimizing the resulting loss or damage to persons and property.
- Principles of safety programme:
  - To identify potential hazards, provide safety equipment and facilities for taking prompt action.
  - To develop safety policies that continuously monitor safety performance and ensure that corrective action is taken when necessary.

- Extent and remedies for accidents:

  Based on the causes the supervisor should apply the art of supervision to control worker's performance to prevent accidents.

  Human failure and machine failure.
- safety training:
  - Systematic training of employees
  - Safety measures should be kept in mind while planning a layout and construction.
  - Workers must be provided with safety tools.
  - Safety posters with slogans at workplaces.
  - National Safety Council.


Safety program evaluation:
Qualitatively: changes in the behavior and attitudes of the employees, changes in the injury-producing conditions.
Quantitatively: incident ratio, severity ratio, and frequency ratio.